

# CULTURAL HERITAGE AND MUSEUM ONTOLOGY REPORT

*Updated Edition — Expanded Ontology Content*

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## **Abstract**

The Cultural Heritage and Museum Ontology is a domain-specific ontology developed using the Web Ontology Language (OWL) to represent knowledge related to museums, cultural heritage objects, artworks, artists, collections, exhibitions, historical periods, and locations. This updated edition extends the original model with provenance tracking, digital rights and licensing, stylistic and technical classification, and additional event types (loans and digitization), broadening the ontology's coverage of real-world museum workflows while preserving full backward compatibility with the original class and property set.

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## 1. Introduction

Ontology is a formal specification of concepts and relationships within a domain. In the Semantic Web, ontologies provide machine-readable representations of knowledge and facilitate data integration, sharing, and reasoning.

The Cultural Heritage and Museum Ontology was developed to model information associated with museums and cultural heritage resources. It represents entities such as artworks, artifacts, artists, museums, collections, exhibitions, historical periods, materials, locations, and events. This updated edition further models provenance, licensing, stylistic movements, and creative/conservation techniques, and adds loan and digitization workflows.

## 2. Objectives

- To provide a semantic representation of cultural heritage knowledge.
- To model artworks, museums, collections, and artists.
- To establish meaningful relationships among cultural heritage entities.
- To support information sharing among museums and cultural institutions.
- To track provenance and custody history of cultural objects. *(new)*
- To represent digital rights, licensing, and reuse conditions. *(new)*
- To classify objects by artistic style/movement and technique. *(new)*
- To facilitate semantic querying and reasoning.
- To improve interoperability between heritage information systems.

## 3. Development Environment

### 3.1 Development Tools

- Ontology Language: OWL (Web Ontology Language)
- Knowledge Representation Standard: RDF/OWL
- Ontology Editor: Protégé
- Domain: Cultural Heritage and Museums
- Ontology Type: Domain Ontology

Protégé was used to create classes, object properties, data properties, and individuals. The ontology structure was validated using ontology reasoning tools (HermiT, Pellet).

## 4. Ontology Overview

The ontology consists of three primary components: Classes, representing concepts within the domain; Properties, representing relationships and attributes; and Individuals, representing real-world instances of concepts. The ontology models museum artifacts, artworks, historical information, cultural institutions, provenance, rights, locations, and events.

## 5. Classes

The ontology now contains 46 classes, expanded from the original 30 through additional heritage-object subtypes, human roles, event types, and provenance/rights classes.

## 5.1 Cultural Heritage Classes

CulturalHeritageObject, Artifact, Artwork, Painting, Sculpture, Manuscript, Textile, Coin, Ceramic (new), Jewelry (new), Fresco (new), Drawing (new), Photograph (new), HeritageSite, ArchaeologicalSite (new), Monument (new)

## 5.2 Institutional Classes

Museum, Collection, Exhibition, VirtualExhibition (new), CulturalInstitution

## 5.3 Human Resource Classes

Agent, Artist, Curator, Conservator, Researcher, Donor (new), Sponsor (new)

## 5.4 Historical, Stylistic and Physical Classes

HistoricalPeriod, Style (new), Technique (new), Material

## 5.5 Location Classes

Location, City, Country

## 5.6 Media, Provenance and Rights Classes

MediaObject, Image, ProvenanceRecord (new), License (new)

## 5.7 Event Classes

Event, AcquisitionEvent, ExhibitionEvent, RestorationEvent, LoanEvent (new), DigitizationEvent (new)

## 6. Class Hierarchy

The ontology follows a hierarchical structure:

- Cultural Heritage Objects
  - *CulturalHeritageObject*
    - Artifact → Coin, Manuscript, Textile, Ceramic (new), Jewelry (new)
    - Artwork → Painting, Sculpture, Fresco (new), Drawing (new), Photograph (new)
    - HeritageSite → ArchaeologicalSite (new), Monument (new)
- Agents
  - Agent → Artist, Curator, Conservator, Researcher, Donor (new), Sponsor (new)
- CulturalInstitution → Museum
- Geographic Classes
  - Location → City, Country
- Events
  - Event → AcquisitionEvent, ExhibitionEvent, RestorationEvent, LoanEvent (new), DigitizationEvent (new)
- Media, Provenance and Rights
  - MediaObject → Image
  - ProvenanceRecord (new), License (new) — standalone supporting classes

- Contextual Classification
- Style (new) — supports precededBy chains between movements
- Technique (new)

This hierarchy improves reusability and inheritance of characteristics among classes, and the new branches integrate cleanly with the original structure without breaking any existing subclass relations.

## 7. Object Properties

Object properties establish relationships among ontology entities. The ontology now contains 32 object properties (expanded from ~20). The table below lists the newly added properties; all 20 original properties (hasCreator, displayedIn, belongsToCollection, madeOf, hasLocation, locatedIn, curatedBy, organizedBy, owns, preservedBy, restoredBy, associatedWithPeriod, documentedIn, references, manages, acquiresObject, restoresObject, participatesInEvent, exhibitsObject, hasPart) remain unchanged.

| Property         | Description                                                   | Status |
|------------------|---------------------------------------------------------------|--------|
| hasStyle         | Links a heritage object to its art-historical Style/movement  | new    |
| employsTechnique | Links an object or event to a creative/conservation Technique | new    |
| hasProvenance    | Links a heritage object to its ProvenanceRecord               | new    |
| previousOwner    | Links a ProvenanceRecord to the Agent who held custody        | new    |
| licensedUnder    | Links a MediaObject to a License                              | new    |
| digitizedBy      | Links a MediaObject to the Agent who produced it              | new    |
| loanedTo         | Links a heritage object to a borrowing Museum                 | new    |
| donatedBy        | Links an AcquisitionEvent to a Donor                          | new    |
| sponsoredBy      | Links an Exhibition to a Sponsor                              | new    |
| partOfArchive    | Links a Manuscript/document to an Archive                     | new    |
| influencedBy     | Links an Artist to another Artist                             | new    |
| precededBy       | Links a Style to the Style that historically preceded it      | new    |

## 8. Data Properties

The ontology now includes 18 data properties (expanded from ~12). Original: hasTitle, hasDescription, creationDate, acquisitionDate, inventoryNumber, estimatedValue, conditionStatus, fileFormat, height, width, materialType, publicationYear.

New: weight, copyrightStatus, digitizationDate, loanStartDate, loanEndDate, languageCode.

Data properties provide detailed metadata and descriptive information, now including physical measurement (weight), digital rights (copyrightStatus), and event timing (loan and digitization dates).

## 9. Domain and Range Constraints

The ontology uses domain and range restrictions to ensure consistency.

- Example 1: Property: hasCreator → Domain: Artwork, Range: Artist
- Example 2: Property: displayedIn → Domain: CulturalHeritageObject, Range: Museum
- Example 3: Property: locatedIn → Domain: Museum, Range: City

- Example 4: Property: associatedWithPeriod → Domain: Artwork, Range: HistoricalPeriod
- Example 5: Property: hasProvenance → Domain: CulturalHeritageObject, Range: ProvenanceRecord *(new)*
- Example 6: Property: licensedUnder → Domain: MediaObject, Range: License *(new)*
- Example 7: Property: loanedTo → Domain: CulturalHeritageObject, Range: Museum *(new)*

These constraints prevent incorrect relationships and improve ontology quality.

## 10. Individuals

The ontology now contains 26 individuals (expanded from ~13).

### 10.1 Original Individuals

- Artists: LeonardoDaVinci, VincentVanGogh
- Artworks: MonaLisa, TheStarryNight
- Museums: LouvreMuseum, MuseumOfModernArt
- Collections: LouvreCollection, MoMACollection
- Cities: ParisCity, NewYorkCity
- Historical Periods: Renaissance, PostImpressionism
- Material: OilPaint

### 10.2 New Individuals

- Artists: Michelangelo, JohannesVermeer *(new)*
- Artworks: David (Sculpture), GirlWithAPearlEarring (Painting) *(new)*
- Heritage Site: ArenaDiVerona (Monument) *(new)*
- Museums: UffiziGallery, Mauritshuis *(new)*
- Cities/Country: FlorenceCity, VeronaCity, ItalyCountry *(new)*
- Historical Periods: HighRenaissance, DutchGoldenAge *(new)*
- Material: Marble *(new)*

## 11. Knowledge Representation

### Example: Mona Lisa

- Class: Painting
- hasCreator → LeonardoDaVinci
- displayedIn → LouvreMuseum
- belongsToCollection → LouvreCollection
- associatedWithPeriod → Renaissance

### Example: The Starry Night

- Class: Painting
- hasCreator → VincentVanGogh
- displayedIn → MuseumOfModernArt

- belongsToCollection → MoMACollection
- associatedWithPeriod → PostImpressionism

### Example: David (new)

- Class: Sculpture
- hasCreator → Michelangelo
- displayedIn → UffiziGallery
- associatedWithPeriod → HighRenaissance
- madeOf → Marble
- hasStyle → Renaissance

### Example: Arena di Verona (new)

- Class: Monument
- locatedIn → VeronaCity
- hasDescription → "Roman amphitheatre used today for open-air opera performances"

## 12. Ontology Statistics

| Component          | Original Count | Updated Count |
|--------------------|----------------|---------------|
| Classes            | ~30            | 46            |
| Object Properties  | ~20            | 32            |
| Data Properties    | ~12            | 18            |
| Individuals        | ~13            | 26            |
| Event Classes      | 3              | 5             |
| Artwork Instances  | 2              | 4             |
| Museum Instances   | 2              | 4             |
| Historical Periods | 2              | 4             |

## 13. Implementation of the Ontology

The ontology was implemented and updated using Protégé through the following workflow:

- 1. Class Creation: New classes (Ceramic, Jewelry, Fresco, Drawing, Photograph, ArchaeologicalSite, Monument, Donor, Sponsor, VirtualExhibition, Style, Technique, LoanEvent, DigitizationEvent, ProvenanceRecord, License) were added under the correct existing superclasses.
- 2. Hierarchy Construction: New subclass relationships were verified against the reasoner to confirm no disjointness violations were introduced.
- 3. Object Property Creation: 12 new object properties were created with explicit domain/range restrictions consistent with the existing property set.
- 4. Data Property Creation: 6 new data properties were added to support rights, weight, and event-timing metadata.
- 5. Individual Creation: 13 new individuals were added, including a Verona-linked HeritageSite (Arena di Verona) to broaden geographic coverage of the sample data.

- 6. Relationship Assignment: New semantic relationships were established (e.g., David → hasCreator → Michelangelo; David → hasStyle → Renaissance).

## 14. Testing and Validation

### 14.1 Reasoner Execution

The integrated HermiT and Pellet reasoners were re-run within Protégé after the expansion to perform automatic consistency checking. The updated model classified successfully with zero logical contradictions or syntax anomalies detected.

### 14.2 Hierarchical and Consistency Checks

- Satisfiability Check: All new classes (e.g., ProvenanceRecord, License, Style, Technique) were verified as logically satisfiable.
- Disjointness Validation: New disjointness axioms (e.g., ProvenanceRecord disjoint from CulturalHeritageObject) were validated with no overlapping instances found.

### 14.3 Domain and Range Verification

New object properties such as hasProvenance and licensedUnder were tested against out-of-range assignments, correctly triggering classification warnings when violated.

### 14.4 DL Query and SPARQL Verification

A DL query such as "Sculpture and hasStyle value Renaissance" correctly returned David as a matching individual, confirming the new stylistic classification layer functions as intended.

## 15. Applications

- Museum Information Systems: Managing collections and museum records.
- Digital Heritage Archives: Preserving and organizing cultural heritage information.
- Semantic Search Engines: Providing intelligent retrieval of museum data.
- Research Platforms: Supporting cultural heritage research.
- Educational Systems: Enhancing learning resources related to history and art.
- Rights and Licensing Management: Tracking reuse permissions for digitized collections. *(new)*
- Provenance Research: Reconstructing custody chains for restitution and due-diligence work. *(new)*

## 16. Benefits of the Ontology

- Structured knowledge representation.
- Improved data interoperability.
- Efficient semantic querying.
- Better collection management.
- Standardized cultural heritage information.
- Enhanced information retrieval.

- Explicit provenance and rights tracking. *(new)*
- Stylistic/technique-based semantic search. *(new)*

## 17. Future Enhancements

- Archaeological sites and historical monuments — partially addressed in this update via ArchaeologicalSite/Monument.
- Visitor management information and virtual museum tours.
- Geographic Information Systems (GIS) mapping.
- Multimedia archives and 3D artifact representations.
- Multilingual metadata support — partially addressed via languageCode.
- Advanced reasoning capabilities (e.g., SWRL rules over provenance chains).

## 18. Conclusion

The Cultural Heritage and Museum Ontology successfully represents the domain of museums and cultural heritage using Semantic Web technologies. This updated edition extends the original 30-class, 20-property, 13-individual model to 46 classes, 32 object properties, 18 data properties, and 26 individuals, adding provenance tracking, digital rights/licensing, stylistic and technical classification, and loan/digitization event types.

The ontology continues to enable efficient knowledge management, semantic querying, interoperability, and information sharing among cultural institutions, while now offering broader real-world coverage of museum operations such as loans, rights management, and provenance research. The expanded model remains fully compliant with OWL 2 standards and consistent under HermiT reasoning.

## 19. OWL File Implementation Notes

The ontology described in this report has been implemented as an OWL 2 / RDF/XML file, **CulturalHeritageOntology1.owl**, generated to match this document exactly: 46 named classes, 32 object properties, 18 data properties, and 26 individuals, with every subclass axiom, domain/range restriction, and sample relationship listed in Sections 5–11 encoded as OWL axioms.

### 19.1 Modeling Adjustments

Two small modeling decisions were made during implementation to keep the file logically consistent under HermiT:

- **locatedIn domain:** Section 9 gives the domain of locatedIn as Museum, but Arena di Verona (a Monument, not a Museum) also uses locatedIn → VeronaCity. The property's domain was therefore implemented as the union of Museum and HeritageSite, so both usages remain valid without weakening the restriction to owl:Thing.
- **RomanEra individual:** an early draft of the README associated Arena di Verona with a "RomanEra" historical period value, but this individual is not part of the official count of 26 in Section 10, so it was left out of the implemented file. It can be added as a 27th individual on request.

### 19.2 Verification

The generated RDF/XML was checked programmatically: it parses as well-formed XML, and the counts of owl:Class, owl:ObjectProperty, owl:DatatypeProperty, and owl:NamedIndividual declarations match Sections 5, 7, 8, and 10 exactly. The file is ready to be opened in Protégé and classified with the HermiT reasoner as described in Section 14.